Optical Instruments

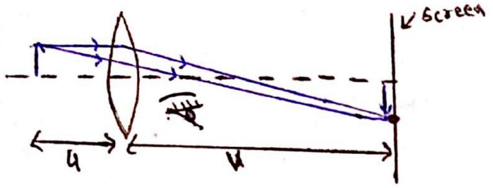
→ Optical instruments are optical devices which help observer to see the micro-detailings of an object, that can't be seen with naked eyes.

Types of Optical Instruments :

Type-01 : Which produce real image of the object.

→ Real images are always observed on a screen.

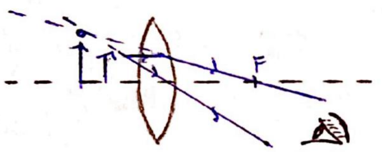
→ In order to observe sharp images, the screen must be placed at the position of image formation.



Similar kind of arrangement is present in 'movie theatre' & 'spectroscopes'.

Type-02 : Which produces virtual image of the object.

→ These optical instruments produce an enlarged image of the object.



In order to observe image, observer must be present in Field of view of the image.

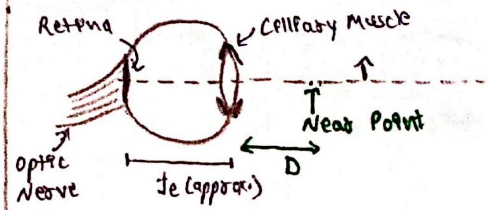
While,
In case of Type-01 optical instruments, image can be observed from any point because it is obtained on screen and screen scatter's light in all dirn.

Human Eye :

→ Eye has a convex lens whose radius of curvature can be accomodated by 'Ciliary muscles'.

→ In order to see clearly, image must be formed at 'Retina' of eye.

→ Distance b/w Retina & eye-lens is approximately focal length of eye lens (f_e)



→ If light coming from ∞, then ciliary muscles are completely relaxed.

→ But as object is bought near eye, ciliary muscles start expanding ($f_e \downarrow$), in order to form image at Retina.

→ It can expand only to a certain limit & f_e can be accomodated only till the object is at 'Near point'

Near Point ⇒

- Minimum distance from eye, upto which f_e can be accomodated & image can be seen with least strain
- Also known as 'Least distance of distinct vision'.

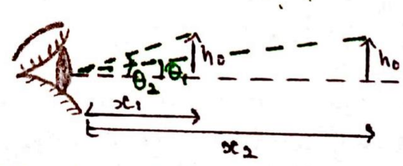
Far Point ⇒

- Max. distance from eye, upto which light can be observed
- For Normal eye ⇒ At ∞
  For Defected eye ⇒ At some finite distance

Angular size of image & object :

→ In case of eye, the actual length & breadth of object does not matter.

→ The perception of size of object for an eye depends on its angular size.



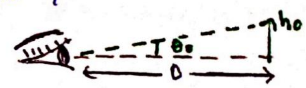
Since angles are small,

$$\theta_1 = \frac{h_o}{x_1} \qquad (\tan\theta \approx \theta)$$

$$\theta_2 = \frac{h_o}{x_2}$$

Simple Microscope :

Base situation ⇒ Object at near point & biggest possible angular size for normal eye.

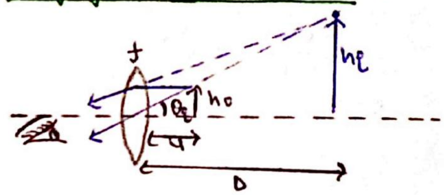


$$\theta_o = \frac{h_o}{D}$$

Simple microscope placed before object ⇒
In order to see more larger image of object, we place a simple microscope b/w eye & object.

→ Object is placed such that image is either formed at Near point (D) or far point (∞).

Image formed at Near Point :→



Here, $\theta_i = \frac{h_o}{u} \Rightarrow \frac{h_o(D+f)}{Df}$

So,

$$m_N = \frac{\theta_i}{\theta_o} = \frac{h_o/u}{h_o/D} \Rightarrow \frac{D\,(D+f)}{Df}$$

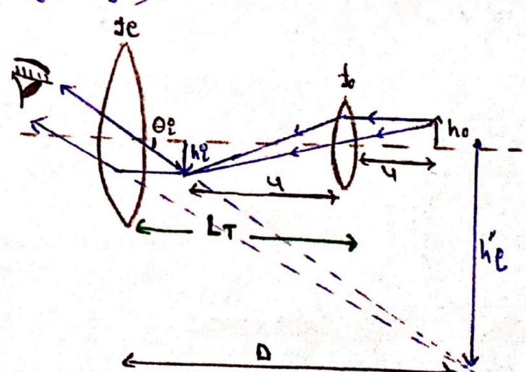
$$\Rightarrow \boxed{m_N = 1 + \frac{D}{f}} \quad \text{(Magnification at Near Point)}$$

$$\boxed{m_F = \frac{h_o/f}{h_o/D} \Rightarrow \frac{D}{f}} \quad \text{(magnification at far point ∞)}$$

→ Image will form at ∞ only when object is kept at F i.e. $u = f$.

Compound Microscope :

→ It contains two convex lenses namely Eye piece & objective lens such that ($f_e > f_o$).



$$\boxed{m_{comp.\,micro.} = m_e \times m_o}$$

When image formed at near point (D) =>

$m_e = 1 + \frac{D}{f_e}$ $m_o = \frac{-v}{u}$

=> $\boxed{m_{CM} = \frac{-v}{u}\left(1+\frac{D}{f_e}\right)}$

When image formed at far point (∞) =>

$m_e = \frac{D}{f_e}$ $m_o = -\frac{v}{u}$

$\boxed{m_{CM} = \frac{-v}{u}\left(D/f_e\right)}$

(*) Tube Length of compound Microscope =>

It is simply the distance b/w eye piece and objective lens.

$L_{TN} = v + u_e$

(when h_i at D) $\boxed{L_{TN} = \left(\frac{u f_o}{u-f_o}\right) + \left(\frac{D f_e}{D+f_e}\right)}$

And,

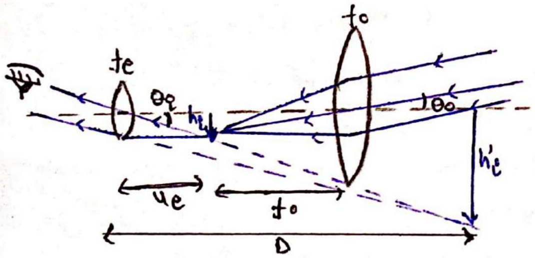
$L_{TF} = f_e + v$

(when h_i at ∞) $\boxed{L_{TF} = \frac{u f_o}{u-f_o} + f_e}$

Refracting Telescope:

It is used to observe the parallel light rays of distant celestial bodies in space.

-> The angle made by parallel beam from very far distance with principal axis is taken as ang. object size.



-> Since light beam are parallel, image by objective lens (hi) will always be at focus (fo).

$\boxed{\theta_o = \frac{h_i}{f_o}}$ $\boxed{\theta_i = \frac{h_i}{u_e}}$

Magnification when image at D =>

$m_N = \frac{h_i/u_e}{h_i/f_o}$ => $\boxed{m_N = \frac{f_o}{f_e}\left(1+\frac{f_e}{D}\right)}$

Magnification when image at ∞ =>

$m_F = \frac{h_i/f_e}{h_i/f_o}$ => $\boxed{m_F = \frac{f_o}{f_e}}$

Tube Length :->

$\boxed{L_{TN} = \frac{D f_e}{D+f_e} + f_o}$ (when image at near point D)

$(u_e + f_o)$

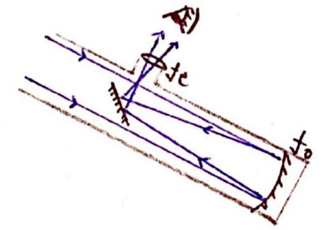
$\boxed{L_{TF} = f_e + f_o}$ (when image at far point ∞)

Reflecting Telescope:

-> It is very cheaper than a refracting telescope.

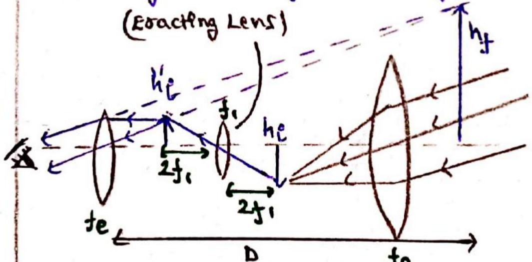
-> Lights are incident on a concave mirror, and there is a plane mirror inclined at 45° at the focus.

-> Reflected light strikes on plane mirror, & travel through eye-piece (convex lens), to the observer's eyes.



Terrestrial Telescope:

-> Same as Refracting Telescope. Only change is that it has an 'Erecting Lens' (Convex) which makes an erect final image instead of inverted.



-> The lenses are placed such that image formed by objective is exactly at centre of curvature (2f1) of erecting lens.

-> Because for object at 2F1 only, image of same size (h'i) would be obtained.

$\boxed{h_i = h_i'}$

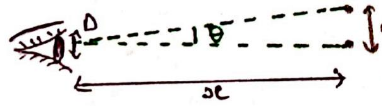
NOTE: In all the relations of optical instruments, put the values with sign.

Limit of Resolution for Optical Instruments:

-> Resolution is defined as how fine the detailing of an image can be observed.

-> In analytical manner, Resolution is ability to observe two points on the image distinctly.

-> For poorly resolved observation, the two points will appear to be infused on each other (i.e. $d \to 0$)



$\theta = \frac{d}{x}$

Limit of Resolution of human eye

$\boxed{\theta = \frac{1.22\lambda}{D}} = \frac{d}{x}$ => $\boxed{x = \frac{dD}{1.22\lambda}}$

Numerical Aperture :->

-> It is the aperture of eye piece required to clearly observe two points having separation of d.

$\boxed{NA = \frac{0.61\lambda}{d}}$